MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

Department of Artificial Intelligence and Machine Learning | Subject Code: AISE309

6-MONTH IMPLEMENTATION PLAN

EvoStrategy: Document Reconciliation Pipeline and Verified Revenue Intelligence for IT Services Enterprises

---

Team Members

- Mazin Moosa (4MT23AI030) — Document Ingestion & OCR Lead
- Adham (4MT23AI032) — Reconciliation Engine Lead
- Ayushi Rastogi (4MT23AI012) — Verification Dashboard Lead
- Prateek M Hulamani (4MT23AI039) — Analytics, Forecasting & Deployment Lead

Under the Guidance of: Mr. Srivatsa Upadhya P, Assistant Professor, Dept. of AI & ML, MITE

1. Project Overview

IT services companies accumulate operational knowledge across thousands of heterogeneous documents — invoices, project reports, client contracts, sprint retrospectives, and financial statements — stored across disconnected formats (PDF, Excel, scanned images) that frequently contain numerical inconsistencies.

EvoStrategy is a four-stage Document Reconciliation Pipeline and Verified Revenue Intelligence System:

- Stage 1 — Document Ingestion using OCR (Tesseract / PaddleOCR)
- Stage 2 — Cross-Document Reconciliation Engine to detect numerical contradictions and entity mismatches
- Stage 3 — Human-in-the-Loop Verification via a Streamlit dashboard
- Stage 4 — Verified Analytics & Forecasting layer, including a What-If simulation module

The system's core innovation is a reconciliation gate that enforces data integrity as a prerequisite to analysis — every projection is traceable to reconciled source documents. The complete system is deployed locally (on-premises), ensuring full data sovereignty and compliance with enterprise confidentiality requirements.

1.1 Implementation Approach

The 6-month plan maps directly onto the project's own four-stage architecture: each stage becomes a primary ownership area for one team member, with monthly integration checkpoints so stages connect continuously rather than only at the end. Three weeks each month are dedicated to focused stage development, and the fourth (or a shared task) is used for cross-stage integration, testing, or deployment work — keeping all four members' progress synchronized.

2. Team & Role Allocation

Member	Primary Ownership	Core Responsibilities
Mazin Moosa (4MT23AI030)	Stage 1: Document Ingestion & OCR	PDF/image preprocessing (PyMuPDF, OpenCV), dual-OCR integration (Tesseract + PaddleOCR), structured JSON field extraction, confidence scoring, ingestion module documentation.
Adham (4MT23AI032)	Stage 2: Cross-Document Reconciliation Engine	Entity resolution (rapidfuzz, spaCy NER), document linking, tolerance-based numerical comparison, discrepancy classification, escalation logic, audit logging.
Ayushi Rastogi (4MT23AI012)	Stage 3: Human-in-the-Loop Verification Dashboard	Streamlit dashboard UI/UX, side-by-side document review, ACCEPT/REJECT/CORRECT workflows, reviewer audit trail, usability testing with non-technical users.
Prateek M Hulamani (4MT23AI039)	Stage 4: Analytics, Forecasting & Deployment	Trend/budget visualizations, regression and ARIMA forecasting, What-If simulation sliders, Docker containerization, Compose orchestration, local 'shoebox' deployment.

Each member also contributes to shared tasks: ground-truth dataset preparation, DFD diagrams, integration testing, and the final report.

3. Technology Stack Summary

Layer	Tools / Libraries
Language & OS	Python 3.11, Ubuntu 22.04 LTS
OCR & Pre-processing	Tesseract 5.x, PaddleOCR 2.x, PyMuPDF (300 DPI rendering), OpenCV (deskew, denoise, contrast)
Entity Resolution	rapidfuzz (Jaro-Winkler ≥ 0.90), spaCy 3.x (NER)
Forecasting	scikit-learn (regression), statsmodels / Prophet (ARIMA time-series)
Dashboard	Streamlit (verification + analytics UI)
Data Tier	SQLite reconciliation DB, local file storage for documents/images, append-only audit logs
Deployment	Docker (per-tier containers), Docker Compose, local-network-only — no internet dependency

4. Six-Month Week-by-Week Plan

The plan is organized into six months of four weeks each (24 weeks), aligned to the four pipeline stages. Shared/integration tasks are marked explicitly in weeks where all members work jointly. Orange rows mark milestone checkpoints.

Month 1 — Foundation, Architecture & OCR Integration

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 1	Repo setup, Docker base image, environment (Python 3.11, Ubuntu 22.04); collect sample invoices, POs, ledgers, payment memos for the dataset	Study reconciliation literature (vendor alias resolution, fuzzy matching); draft document-relationship rules (invoice ↔ PO ↔ memo ↔ ledger)	Wireframe the Streamlit verification dashboard; define reviewer roles and session/auth flow	Set up SQLite schema for extraction DB, reconciliation registry, verified store; draft Docker Compose skeleton
Wk 2	Finalize per-transaction JSON schema {vendor, date, amount, category, document_type, source_file, confidence_score}; benchmark Tesseract vs PaddleOCR on sample docs	Build vendor alias dictionary; prototype entity resolution with rapidfuzz (Jaro-Winkler) and spaCy NER	Build Streamlit app skeleton; static mockups for record-review and analytics screens	Evaluate forecasting libraries (Prophet vs statsmodels/ARIMA); design data-tier file/DB layout
Wk 3	Implement PDF rendering (PyMuPDF, 300 DPI) and image preprocessing (OpenCV: deskew, denoise, contrast normalization)	Continue entity resolution prototype; define date-window + vendor-matching logic for document linking	Build dashboard navigation shell; integrate placeholder data into review screen	Define budget configuration schema (total budget, period, category allocations); plan what-if parameter set
Wk 4	Integrate Tesseract (printed text) + PaddleOCR (tables/layouts); attach confidence scoring per field	Finalize relationship-rule config (YAML) for document linking	Connect dashboard shell to mock reconciliation registry for layout testing	Set up CI scaffold and base Docker image for processing containers
MILESTONE 1 — Architecture, schemas, and tooling finalized; OCR engines integrated				

Month 2 — Stage 1 Completion & Stage 2 Initiation

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 5	Complete field extraction to structured JSON; implement low-confidence flagging (< 0.85)	Implement document linking (Phase 1): vendor similarity + date proximity matching across documents	Build side-by-side source-document viewer component for the dashboard	Containerize the OCR/ingestion module; write integration tests for Stage 1 output format

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 6	Test Stage 1 against ≥ 92% field-accuracy target; tune OCR + preprocessing on failure cases; produce per-document processing log	Implement numerical comparison engine with tolerance bands (default 2%); auto-resolution of in-tolerance discrepancies	Design ACCEPT / REJECT / CORRECT action UI with reviewer-identity capture	Build amount and date normalization utilities (currency parsing, GST separation, multi-format date parsing) shared across stages
Wk 7	Support handoff of Stage 1 output to Stage 2; fix edge cases in scanned/low-quality documents	Implement escalation-flag generation for discrepancies beyond tolerance or missing entries	Wire dashboard to live (non-mock) reconciliation registry; render discrepancy magnitude and type per record	Set up Docker networking between processing, data, and presentation tiers (Compose draft v1)
Wk 8	Assist in building a labeled ground-truth dataset for reconciliation testing	Implement discrepancy-type classification (numerical mismatch / missing document / entity alias / duplicate entry) and audit-log entries	Complete MVP of verification dashboard (view + flag inspection, no actions yet)	Review Stage 1+2 integration; begin Stage 4 data-model design (reconciled record store schema)
MILESTONE 2 — Stage 1 (Ingestion & OCR) finalized and frozen; Stage 2 prototype functional				

Month 3 — Stage 2 Completion & Stage 3 Initiation

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 9	Support Stage 2 testing with extraction edge cases; document Stage 1 module (inference/extraction interface)	Run precision/recall evaluation of reconciliation engine against ground truth; tune fuzzy-match thresholds	Implement ACCEPT / REJECT / CORRECT actions with full audit logging (reviewer identity + timestamp)	Begin Stage 4 module: spending trend visualization (monthly/quarterly, by vendor and category)
Wk 10	Cross-validate OCR confidence scores against reconciliation escalations to tune the 0.85 threshold	Refine auto-resolution vs escalation boundary; reduce false-positive escalation rate	Implement quarantine flow for rejected records (with reason) and correction-record storage (original/corrected/reviewer fields)	Build category-level budget consumption gauge charts
Wk 11	Begin end-to-end Stage 1→2 integration testing	Stage 2 regression testing; finalize reconciliation status output (MATCHED / AUTO-RESOLVED / ESCALATED / MISSING)	Usability pass on dashboard with a non-technical-user persona; add contextual explanations for reviewers	Implement linear-regression runway projection with 85% confidence interval
Wk 12	Support full pipeline smoke test (Stage 1 → 2 → 3 mock)	Finalize Stage 2 documentation and audit-log format	Polish dashboard UI (layout, contrast, navigation) ahead of Milestone 3 review	Implement ARIMA time-series forecast (3–6 month horizon)
MILESTONE 3 — Stage 2 (Reconciliation Engine) finalized; Stage 3 (Verification Dashboard) MVP complete				

Month 4 — Stage 3 Completion, Stage 4 Initiation & First Full Integration

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 13	Optimize OCR throughput for batch processing (multi-page documents)	Support Stage 3 integration — verified record promotion from reconciliation registry	Connect verified-record promotion to reconciled record store; finalize audit trail end-to-end	Connect Stage 4 visualizations to the reconciled record store (real data, not mocks)
Wk 14	Joint task: wire Stage 1 → Stage 2 → Stage 3 data flow end-to-end	Joint task: validate reconciliation outputs flowing correctly into dashboard	Joint task: verify dashboard correctly reflects live reconciliation state; fix integration bugs	What-if simulation module: real-time Streamlit sliders (vendor consolidation, headcount change, pricing adjustment)
Wk 15	Write Stage 1 unit tests; measure OCR accuracy on full sample set	Write Stage 2 unit tests; measure precision/recall on full sample set	Write Stage 3 usability/functional tests; capture reviewer-decision audit trail samples	Recompute-on-slider-change logic for what-if module; validate against verified historical data
Wk 16	Bug-fix pass from Stage 1 integration tests	Bug-fix pass from Stage 2 integration tests	Bug-fix pass from Stage 3 integration tests; finalize dashboard navigation for analytics module	Annotate all charts with data-source traceability (record count, date range, verification status)
MILESTONE 4 — Full pipeline (Stages 1→2→3→4) wired end-to-end; What-if module functional				

Month 5 — Stage 4 Completion, What-If Module & Containerized Deployment

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 17	Write OCR/ingestion module documentation and developer notes	Write reconciliation engine documentation; tune entity-resolution edge cases found in integration	Integrate analytics views into the main dashboard navigation	Build what-if comparison table (baseline vs scenario projections)
Wk 18	Dockerize ingestion module as an isolated processing container	Dockerize reconciliation engine as an isolated processing container	Dockerize Streamlit presentation container	Finalize Docker Compose orchestration (processing + data + presentation tiers)
Wk 19	Joint task: test 'shoebox' single-workstation deployment (all containers on one machine, no internet)	Joint task: verify local-network-only operation; confirm no external calls	Joint task: validate dashboard accessibility from a separate client device on the local network	Lead deployment integration; resolve container networking and volume-mount issues
Wk 20	Support deployment smoke tests on Stage 1	Support deployment smoke tests on Stage 2	Support deployment smoke tests on Stage 3	Run full-system smoke test; fix deployment-blocking issues
MILESTONE 5 — Complete system deployed locally end-to-end ('shoebox' model); all stages verified post-deployment				

Month 6 — Evaluation, Hardening, Documentation & Final Report

Week	Mazin Moosa — Ingestion & OCR	Adham — Reconciliation Engine	Ayushi Rastogi — Verification Dashboard	Prateek M Hulamani — Analytics, Forecasting & Deployment
Wk 21	Performance/load testing: batch OCR throughput under concurrent document loads	Security pass: access roles and tamper-evident audit log review for reconciliation engine	Security pass: encrypted storage and role-based access for dashboard sessions	Security pass: data-tier encryption at rest; review audit-log immutability
Wk 22	Run OCR accuracy evaluation vs 80% baseline target (≥ 92% goal)	Run reconciliation precision/recall evaluation (≥ 0.90 / ≥ 0.88 goals)	Measure human-review-efficiency improvement vs fully manual baseline (≥ 60% reduction goal)	Run forecast-accuracy evaluation (≤ 15% MAPE goal) against held-out periods
Wk 23	Fix issues found in Stage 1 evaluation; finalize DFD Level 2 detail for ingestion	Fix issues found in Stage 2 evaluation; finalize DFD Level 2 (reconciliation engine detail)	Fix issues found in Stage 3 evaluation; write end-user manual for finance reviewers	Fix issues found in Stage 4 evaluation; finalize DFD Level 0/Level 1 diagrams
Wk 24	Contribute Stage 1 section to final project report	Contribute Stage 2 section to final project report	Contribute Stage 3 section to final project report; coordinate report formatting	Contribute Stage 4 + deployment sections to final project report; consolidate evaluation results

5. Milestone Summary

ID	Timing	Deliverable	Owner(s)
M1	End of Month 1	Architecture, schemas, dataset, and tooling finalized; OCR engines integrated	All
M2	End of Month 2	Stage 1 (Ingestion & OCR) frozen and tested; Stage 2 (Reconciliation) prototype functional	Mazin Moosa (4MT23AI030) (lead), Adham (4MT23AI032) (lead)
M3	End of Month 3	Stage 2 finalized with precision/recall validation; Stage 3 (Dashboard) MVP complete	Adham (4MT23AI032) (lead), Ayushi Rastogi (4MT23AI012) (lead)
M4	End of Month 4	Full pipeline (Stage 1→2→3→4) wired end-to-end; What-if simulation functional	All
M5	End of Month 5	Complete system containerized and deployed locally in the 'shoebox' single-workstation model	Prateek M Hulamani (4MT23AI039) (lead), All support
M6	End of Month 6	System evaluated against all hypotheses; documentation, audit, and final report complete	All

6. Evaluation Metrics & Hypotheses

Drawn directly from the project's own hypotheses and evaluation criteria, validated in Month 6 against a ground-truth dataset:

Metric	What It Measures	Target	Owner
OCR & Extraction Accuracy	Field-level extraction accuracy on financial documents	≥ 92% (vs. 80% baseline)	Mazin Moosa (4MT23AI030)
Reconciliation Precision / Recall	Correct identification of genuine discrepancies	≥ 0.90 precision, ≥ 0.88 recall	Adham (4MT23AI032)
Forecast Accuracy	Mean Absolute Percentage Error of budget/runway projections	≤ 15% MAPE	Prateek M Hulamani (4MT23AI039)
Human Review Efficiency	Reduction in manual reconciliation effort vs. fully manual baseline	≥ 60% reduction	Ayushi Rastogi (4MT23AI012)

7. Risk Management

Risk	Likelihood	Impact	Mitigation
OCR accuracy degrades on poor-quality scans / handwriting	Medium	High	Preprocessing tuning (deskew/denoise); dual-engine fallback (Tesseract + PaddleOCR); manual-correction path via Stage 3 dashboard
Fuzzy-matching produces false-positive vendor links	Medium	Medium	Tune Jaro-Winkler threshold (≥ 0.90) using ground-truth validation; allow reviewer override in Stage 3

Risk	Likelihood	Impact	Mitigation
Stage integration delays (Stage 2 depends on Stage 1 output format)	Medium	High	Freeze the per-transaction JSON schema in Week 2; integration checkpoints at the end of every month, not just at milestones
Docker networking/orchestration issues at deployment	Low	Medium	Early Compose draft in Month 2; dedicated integration week (Wk 19) before final evaluation
Forecast MAPE target missed due to limited historical data	Medium	Medium	Use ARIMA + linear regression ensemble; clearly bound confidence intervals; document data limitations in report
Uneven workload near final submission	Medium	Medium	Joint integration tasks scheduled every month (not just Month 6); weekly stand-up to rebalance early

8. Final Deliverables Checklist

- Working end-to-end pipeline: Ingestion → Reconciliation → Verification → Analytics/Forecasting
- Streamlit dashboard supporting human-in-the-loop review with full audit trail
- Dockerized, locally-deployable system (single-workstation 'shoebox' model)
- Evaluation report against all four hypotheses (OCR accuracy, reconciliation precision/recall, forecast MAPE, review-efficiency gain)
- DFD Level 0, 1, and 2 diagrams matching the implemented system
- End-user manual for finance reviewers and a developer/module reference
- Final project report and viva/demo presentation